

Pharmacology of agents used for Blood disorder

Introduction

Hematopoiesis, is the productions of circulating **erythrocytes**, **platelets**, and **leukocytes** from undifferentiated stem cells.

>**200** billion new blood cells per day in the normal person .

- The **hematopoietic machinery** resides primarily in the bone marrow in adults and requires a constant supply of three essential nutrients—
 - Iron
 - Vitamin B12,
 - and folic acid
- Inadequate supplies of either the essential nutrients or the **growth factors** result in deficiency of **functional blood cells**

Anemia

- **Anemia**, is a deficiency in oxygen-carrying erythrocytes
-it is abnormally low hemoglobin concentration in the blood

Clinical presentation

- acute blood loss.....hypovolemia
- Signs of vascular instability appear with acute losses of **10–15%** of the total blood volume.
- signs of hypovolemic shock including confusion, dyspnea, diaphoresis, hypotension.

Classification

Type	MCV	Common Causes
Macrocytic	Increased	Folic acid deficiency Vitamin B ₁₂ deficiency Liver disease Alcohol Hypothyroidism Drugs (sulfonamides, zidovudine, antineoplastic agents) Myelodysplastic syndromes
Microcytic	Decreased	Iron deficiency Thalassemias
Normocytic	Normal	Aplastic anemia Anemia of chronic disease Chronic kidney disease Hemolytic anemia Spherocytosis

Drugs used in anemia

Iron and its pharmacology-

- Iron deficiency is the most common cause of chronic anemia.
- Fatigue, dizziness, exertional dyspnea, and other generalized symptoms of tissue **hypoxia**.
- Hemoglobin reversibly binds **oxygen** and provides the critical mechanism for **oxygen delivery** from the lungs to other tissues.
- In the absence of adequate **iron**, small **erythrocytes** with insufficient **hemoglobin** are formed
- **Iron-containing** heme is a component of
 - **myoglobin**,
 - **cytochromes**, and other proteins with diverse biologic functions.

Sources of Iron

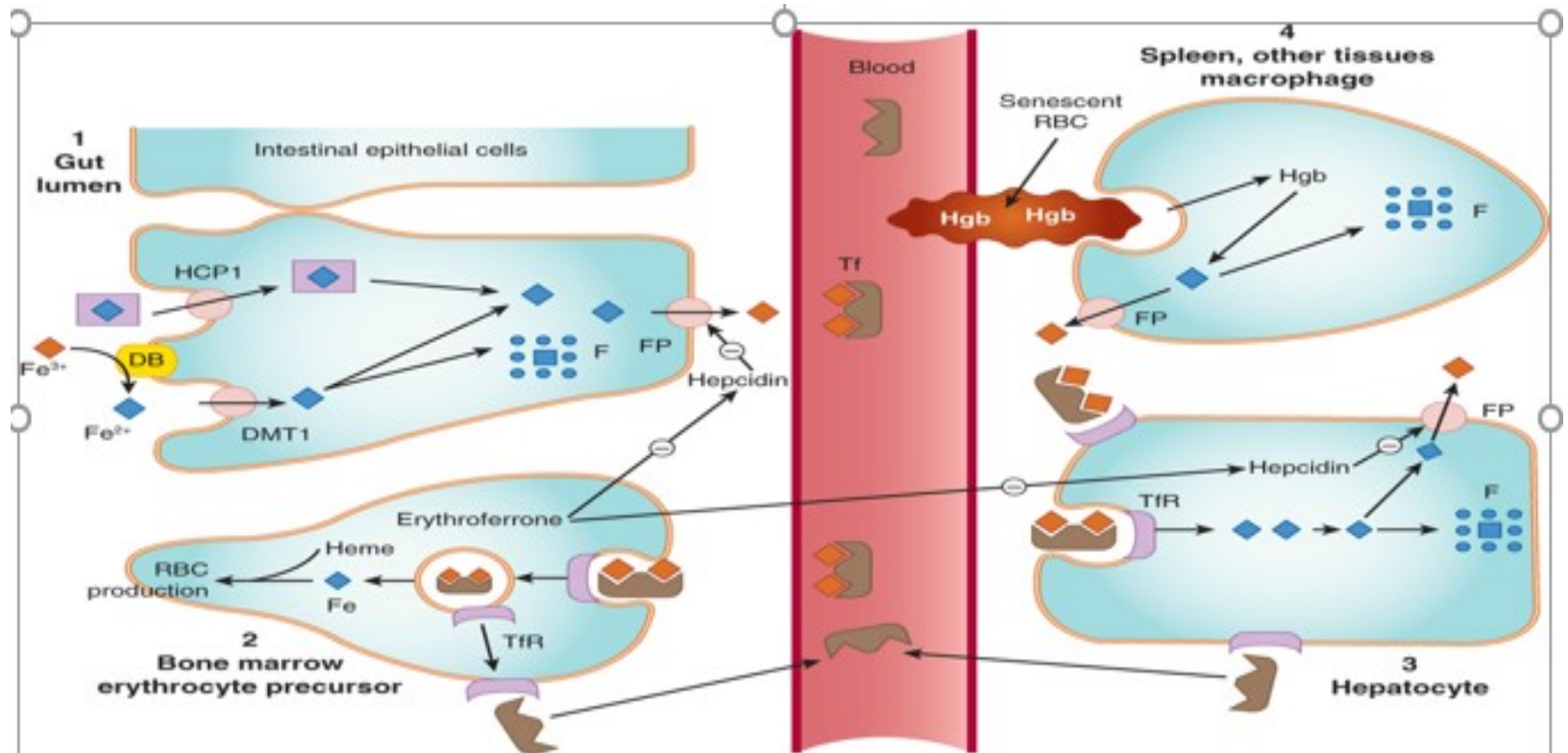
- Iron is available in a wide variety of foods but is especially abundant in **meat**.
- The iron in **meat** can be efficiently absorbed, because
 - heme iron in meat hemoglobin and myoglobin can be absorbed intact without first having to be dissociated into elemental iron
- **Iron** in vegetables and grains, is often less absorbed
 - tightly bound to organic compounds

Pharmacokinetics

- Of **10–15 mg**, A normal individual absorbs **5–10%** of this iron, or about **0.5–1 mg** daily.
- **Sites of absorption**
 - ✓ duodenum and
 - ✓ proximal jejunum
- Iron absorption increases in response to **low iron stores** or increased **iron requirements**.
- Total iron absorption
 - 1–2 mg/d in menstruating women
 - 3–4 mg/d in pregnant women.

Absorption...

- Nonheme iron in foods and iron in inorganic iron salts and complexes must be reduced by a **ferrireductase** to ferrous iron (Fe^{2+}) before it can be absorbed by intestinal mucosal cells.



Transport

- ❖ Iron is transported in the plasma bound to **transferrin, a β -globulin** that can bind **two molecules of ferric iron**
- ❖ The transferrin-iron complex enters maturing erythroid cells by a **specific receptor mechanism**.
- ❖ erythroid cells—bind and internalize the transferrin-iron complex through the process of **receptor-mediated endocytosis**.

- ❖ In the cytoplasm, it is funneled into hemoglobin synthesis or stored as **ferritin**
- ❖ The **transferrin-transferrin** receptor complex is recycled to the **cell membrane**, where the transferrin dissociates and returns to the **plasma**
- ❖ **Iron store depletion** and iron deficiency anemia are associated with an increased concentration of serum **transferrin**

Storage

- ❖ Iron is stored in
 - intestinal mucosal cells,
 - stored, primarily as ferritin,
 - in the **liver, spleen, and bone,**
- ❖ The mobilization of iron from macrophages and hepatocytes is primarily controlled by **hepcidin regulation** of ferroportin activity.

Elimination

- ❖ There is no mechanism for excretion of iron.
- ❖ Small amounts are lost in the feces **by exfoliation** of intestinal mucosal cells,
- ❖ **trace amounts are excreted in bile, urine, and sweat.**
 - These losses account for no more than **1 mg** of iron per day.
- ❖ body's ability to excrete iron is so limited,
 - regulation of iron balance is achieved through changing **intestinal absorption and storage of iron in response to the body's needs.**

Clinical Pharmacology

Indications for the Use of Iron

- indication for the use of iron preparations is the **treatment or prevention of iron deficiency anemia.**
 - For **hypochromic, microcytic anemia** in which the **erythrocyte mean cell volume (MCV)** and the **mean cell hemoglobin concentration are low**
- Iron deficiency is commonly seen in populations with increased iron requirements.
 - infants, especially premature infants;
 - children during rapid growth periods;
 - pregnant and lactating women; and
 - patients with chronic kidney disease

- The most common cause of iron deficiency in adults is blood loss.
 - Menstruating women lose about **30 mg** of iron with each menstrual period

Treatment

- Iron deficiency anemia is treated with
 - oral or
 - parenteral iron preparations.
- Oral iron corrects the anemia just as rapidly and completely as **parenteral iron** with **normal GI absorption**
- Parenteral is required
 - advanced chronic kidney (hemodialysis) and treatment with **erythropoietin**.

Oral iron therapy

- ❖ A wide variety of **oral iron** preparations is available. Because ferrous iron is absorbed,
 - Ferrous sulfate,
 - ferrous gluconate and
 - ferrous fumarate
- ❖ Different iron salts provide different amounts of elemental iron

Preparation	Tablet Size	Elemental Iron per Tablet	Usual Adult dosage
Ferrous sulfate	325 mg	65 mg	2–4
Ferrous sulfate, desiccated	200 mg	65 mg	2–4
Ferrous gluconate	325 mg	36 mg	3–4
Ferrous fumarate	325 mg	106 mg	2–3

Adverse effects

- Common adverse effects of oral iron therapy include
 - nausea,
 - epigastric discomfort,
 - abdominal cramps,
 - constipation, and diarrhea.
- These effects are usually dose-related
 - overcome by lowering the daily dose of iron or
 - by taking the tablets immediately after or with meals.
- Patients taking oral iron develop **black stools**.

Parenteral iron therapy

- ❖ Parenteral therapy is required
 - iron deficiency who are unable to tolerate oral
- ❖ This includes patients with
 - advanced chronic renal disease
 - postgastrectomy conditions and previous small bowel resection,
 - inflammatory bowel disease involving the proximal (malabsorption)

Parenteral iron therapy...

❖ The challenge with parenteral iron therapy

- parenteral administration of inorganic free ferric iron produces serious dose-dependent toxicity,
 - **severely limits** the dose that can be administered.
- Therefore
 - **formulated as a colloid containing** particles with a core of iron oxyhydroxide surrounded by a core of carbohydrate

❖ There are three forms of parenteral Fe preparation

- iron dextran,
- sodium ferric gluconate complex, and
- iron sucrose

Clinical Toxicity

Acute Iron Toxicity

- As few as 10 tablets of (any form of preparation)
 - lethal in young children.
 - instructed to store tablets in child-proof containers out of the reach of children.
- **Whole bowel irrigation** should be performed to flush out unabsorbed pills.
- **antagonist (Deferoxamine)....**Should be given IV
 - to bind iron that has already been absorbed and to promote its excretion in urine and feces.
- **Activated charcoal**, a highly, does *not* bind iron and thus is ineffective.
- **supportive therapy** for gastrointestinal bleeding, metabolic acidosis, and shock must also be provided

Chronic Iron Toxicity

- **Hemochromatosis is chronic toxicity due to Fe overload**
 - excess iron is deposited in the heart, liver, pancreas, and other organs.
 - It can lead to organ failure and death.
- Occurs mainly in patients with
 - inherited hemochromatosis, and
 - in patients who receive many red cell transfusions (eg, individuals with β -thalassemia).
- Chronic iron overload is most efficiently treated by
 - intermittent phlebotomy or
 - Iron chelation therapy (parenteral or oral)

Vitamin b12

- ❖ Vitamin B12 (cobalamin) serves as a cofactor for several **essential biochemical reactions**
- ❖ Deficiency leads to megaloblastic anemia
- ❖ deficiency of vitamin B12 is due to due to
 - inadequate absorption of dietary vitamin B12 (**most common**) an
 - inadequate supply in the diet

Clinical features of B12 deficiency

- The most characteristic clinical manifestation of vitamin B₁₂ deficiency is **megaloblastic, macrocytic anemia**.
- The neurologic syndrome associated with vitamin B₁₂ deficiency usually begins with
 - **paresthesias** in peripheral nerves and
 - weakness and progresses to **spasticity, ataxia**, and other central nervous system dysfunctions.
- Correction of vitamin B₁₂ deficiency arrests the progression of neurologic disease (**not fully reverse it**)

Pharmacokinetics

- ❖ In a diet contains 5–30 mcg of vitamin B₁₂ daily, **1–5 mcg** is usually absorbed.
- ❖ stored, primarily in the **liver**,
 - In adult there is a total vitamin B₁₂ storage pool of 3000–5000 mcg.
- ❖ Only trace amounts of **vitamin B₁₂** are normally lost in urine and stool.
- ❖ normal daily requirements of vitamin B₁₂ are only about **2 mcg**,
- ❖ It would take about **5 years** for all of the stored vitamin B₁₂ to be exhausted and for megaloblastic anemia to develop if **B₁₂** absorption were stopped.

Absorption and Transport

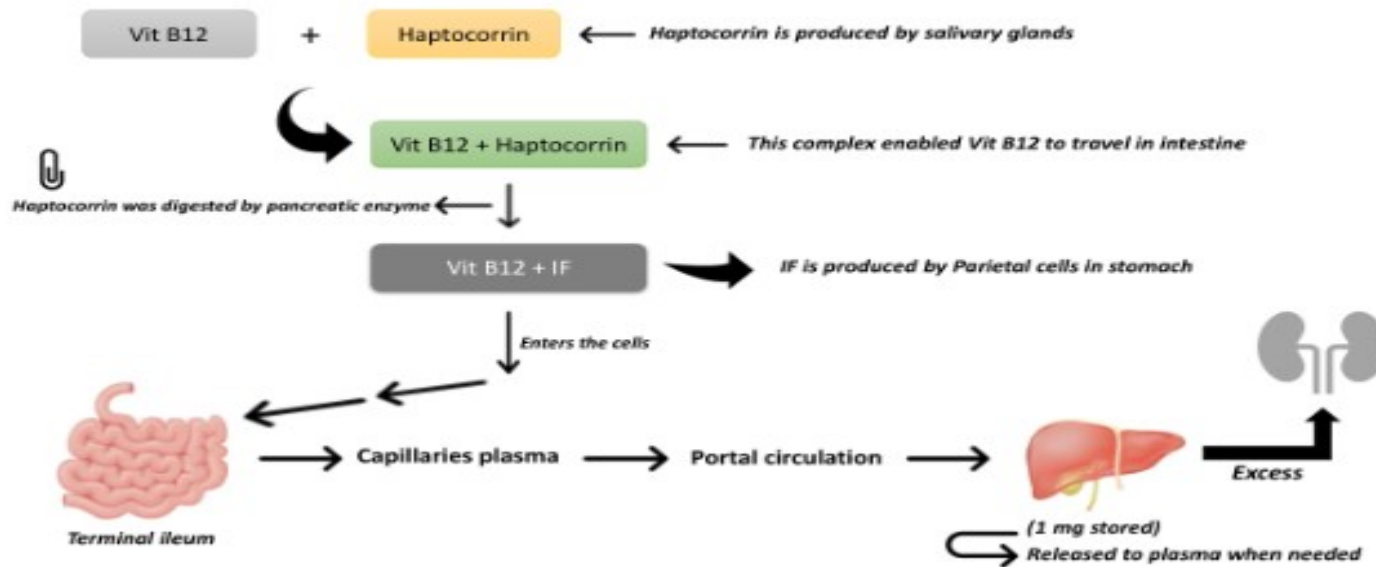


Figure: Vitamin B12 absorption and transport

Absorption

- ❖ Vitamin B₁₂ will be absorbed after it complexes with intrinsic factor,
- ❖ **intrinsic factor–vitamin B₁₂ complex** is subsequently absorbed in the distal **ileum** by a highly selective **receptor-mediated transport system**.
- ❖ deficiency in humans most often results from malabsorption of **vitamin B₁₂** due
 - either to lack of intrinsic factor or to
 - loss or malfunction of the absorptive mechanism in the distal ileum.
- ❖ **Transportation**
 - After absorption vitamin B₁₂ is **transported** to the various cells of bound to glycoproteins, **transcobalamin I, II, and III**.
 - **Excess vitamin B₁₂** is stored in the liver

Interrelationships and metabolic roles of vitamin B12 and folic acid

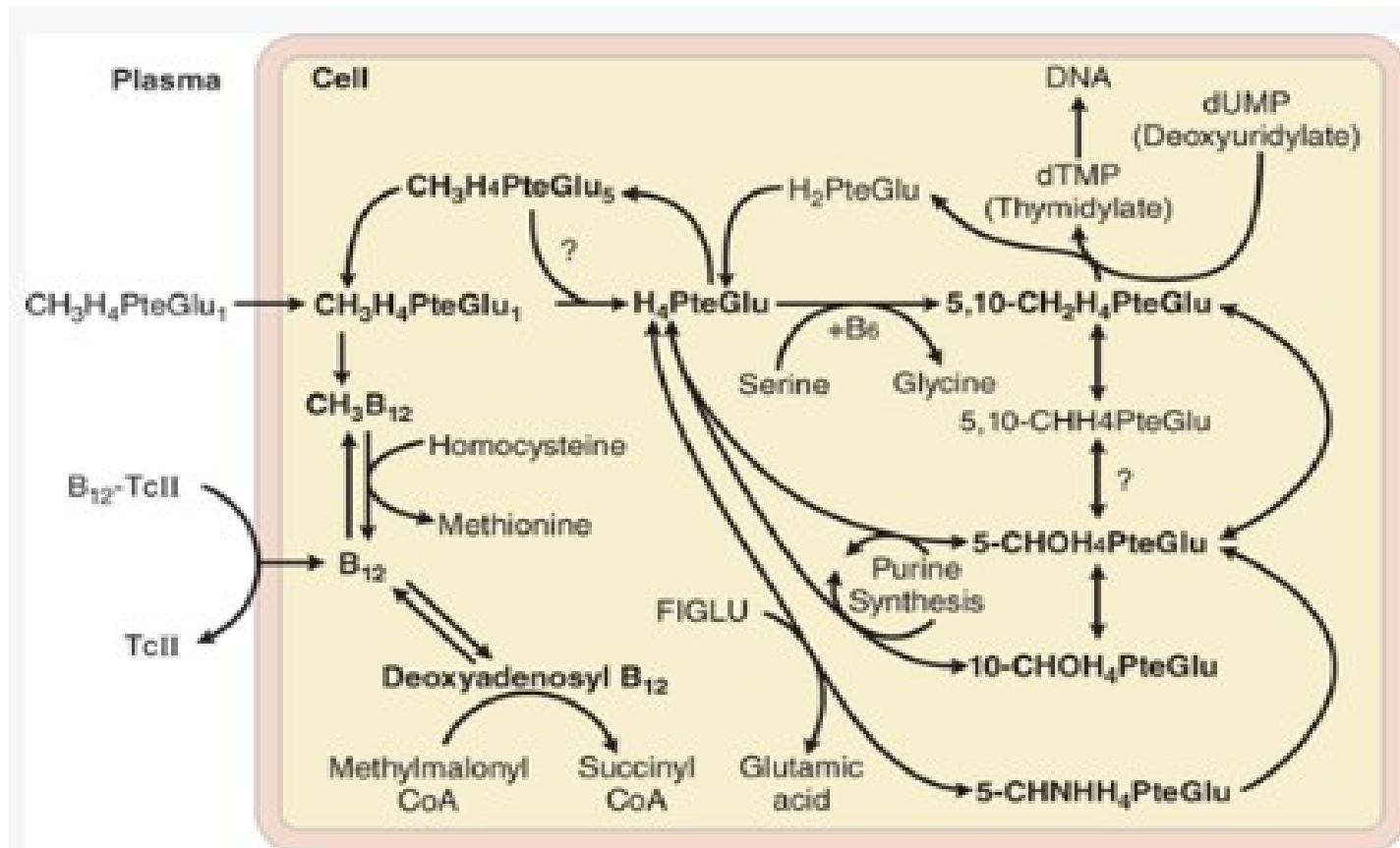


Figure: Interrelationships and metabolic roles of vitamin B12 and folic acid

Diagnosis

- ❖ Once a diagnosis of megaloblastic anemia is made,
 - it must be determined whether vitamin B₁₂ or folic acid deficiency is the cause. Because of
 - (Other causes of megaloblastic anemia are very rare.)
- ❖ Diagnosis accomplished by
 - measuring serum levels of the [vitamins](#).
 - Normal level ~200 to 900 pg/m
- Level of MMA

Diagnosis...

- ❖ partial or total gastrectomy, and
- ❖ Conditions that affect the **distal ileum**, such as malabsorption syndromes,
 - inflammatory bowel disease, or
 - small bowel resection.
- Strict vegans eating a diet free of meat and dairy products may become B₁₂ deficient

❖ **Pernicious anemia**

- results from defective secretion of **intrinsic factor** by the gastric mucosal cells.

❖ Patients with pernicious anemia

- have gastric atrophy and

- fail to secrete intrinsic factor (as well as hydrochloric acid).

- These patients frequently have autoantibodies to intrinsic factor.

❖ Vitamin B₁₂ deficiency also occurs when the region of the **distal ileum** that absorbs the

- vitamin B₁₂–intrinsic factor complex is damaged, as when the **ileum** is involved with **inflammatory bowel disease** or when the ileum is **surgically resected**.

- Rare cases of **vitamin B₁₂ deficiency** in children have been found to be
 - secondary to **congenital deficiency** of intrinsic factor or to
 - defects of the **receptor sites** for vitamin B₁₂—intrinsic factor complex located in the distal ileum.

Management of deficiency

- ❖ Since most vitamin B₁₂ deficiency are caused by malabsorption of the vitamin;
 - **parenteral injections of vitamin B₁₂** are required for therapy.
- ❖ **the underlying disease** should be treated after initial treatment with parenteral vitamin B₁₂.
- ❖ **lifelong treatment** with may be vitamin B₁₂ required .

Management of deficiency

Vitamin B₁₂ for parenteral injection is available as

- cyanocobalamin or
- hydroxocobalamin.

❖ Hydroxocobalamin is preferred because

- it is more highly protein-bound and therefore remains longer in the circulation.
- Initial therapy should **100–1000 mcg** of vitamin B₁₂ IM/day
- intramuscularly daily or every other day for 1–2 weeks to replenish body stores.
- **Maintenance therapy** consists of 100–1000 mcg intramuscularly once a month for life.

Management of deficiency

- ❖ If neurologic abnormalities are present,
 - maintenance therapy injections every **1–2** weeks for **6 months** before switching to monthly injections.
- ❖ oral doses of **1000 mcg** of vitamin B₁₂ daily to treat patients with pernicious anemia who refuse parenteral injection

Folic acid

- ❖ Reduced forms of folic acid are required to provide precursors for the synthesis of
 - amino acids,
 - purines, and
 - DNA.
- ❖ folate deficiency is implicated
 - Congenital **malformations** in newborns
- ❖ Folate deficiency is easily corrected by administration of **folic acid**.

❖ Folic acid deficiency caused by

- Inadequate dietary **intake of folates**.
- Patients with **alcohol dependence** and patients with **liver disease**
- Pregnant women
- hemolytic anemia
- Patients with malabsorption syndromes also frequently develop **folic acid** deficiency.
- Patients who require renal dialysis

Drug induced folic acid deficiency

❖ Folic acid deficiency can be caused by drugs.

- Methotrexate
- Trimethoprim
- pyrimethamine
- phenytoin

Pharmacokinetics & source

- ❖ The average American diet contains **500–700** mcg of folates daily,
 - **50–200** mcg of which is usually absorbed, depending on metabolic requirements.
- ❖ folic acid is readily and completely absorbed in the **proximal jejunum**
- ❖ **In pregnant woman absorption**
 - 300–400 mcg of folic acid daily.
- ❖ **Sources**
 - Vegetables
 - and **animal tissues**; Yeast, Liver & kidney

Pharmacokinetics

❖ Excretion

- ❖ excreted in the **urine** and **stool** and are also destroyed by catabolism.
- ❖ **body stores of folates are relatively low and daily requirements high,**
 - Folic acid deficiency and **megaloblastic anemia** can develop within **1–6 months** after the intake of folic acid stops,

Clinical Pharmacology

❖ Megaloblastic anemia **due to folate**

- microscopically indistinguishable from the anemia caused by vitamin **B₁₂ deficiency**
- does not cause the characteristic **neurologic syndrome seen in vitamin B₁₂ deficiency.**

Management of deficiency

- ❖ **Parenteral administration of folic acid is rarely necessary**
 - **oral folic acid** is well absorbed even in patients with **malabsorption syndromes**.
- ❖ **1 mg folic acid orally daily is sufficient to**
 - reverse megaloblastic anemia,
 - restore normal serum folate levels, and
 - replenish body stores of folates in almost all patients.

Management of deficiency

❖ Folic acid supplementation is required to prevent **folic acid deficiency** should be considered in high risk patients

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